

Lab 01 - The Price of One Request

Provider: Google Gemini API

Model: gemini-3.6-flash

Languages: English, Russian, Kazakh

Group: 6

1. Introduction

The main goal of this laboratory work was to see how the same text behaves in different languages and how this affects the number of tokens and the final cost of an API request. I worked with English, Russian and Kazakh. I also compared two tokenizers: o200k_base and cl100k_base.

For the practical API part, I used the Google Gemini API and the model gemini-3.6-flash. First I checked the text locally without making API requests. Then I measured characters, UTF-8 bytes and tokens. After that, I made real API requests and calculated the approximate cost for a larger number of requests.

2. Part 0 - Tokenizer comparison

First, I ran the tokenizer script. This part did not require an API key because everything was calculated locally.

Text	o200k_base	cl100k_base
Sentence - EN / RU / KK	12 / 18 / 21	12 / 35 / 58
Complaint - EN / RU / KK	59 / 80 / 118	59 / 146 / 265

For the sentence, o200k_base gave $1.50\times$ Russian/English and $1.75\times$ Kazakh/English. With cl100k_base the ratios were $2.92\times$ and $4.83\times$. For the complaint, the ratios were $1.36\times$ and $2.00\times$ for o200k_base, and $2.47\times$ and $4.49\times$ for cl100k_base.

My observation was that the tokenizer can make a big difference. In my data, o200k_base was much more efficient for Russian and Kazakh than cl100k_base.

3. Part 1 - Offline analysis and prediction

Next, I ran the offline analysis before making the real API request. For the complaint I got 300 characters / 300 bytes / 54 words for English, 315 characters / 576 bytes / 45 words for Russian, and 344 characters / 640 bytes / 42 words for Kazakh.

My prediction was Russian/English $\approx 2.0\times$ and Kazakh/English $\approx 2.2\times$. I based this mainly on UTF-8 bytes because Russian and Kazakh use Cyrillic characters.

The actual tokenizer results were $1.36\times$ and $2.00\times$ for Russian and Kazakh with o200k_base, and $2.47\times$ and $4.49\times$ with cl100k_base. So my prediction was reasonably close for o200k_base, but cl100k_base produced a much larger difference.

This showed me that the number of bytes alone cannot fully predict the number of tokens. The tokenizer's vocabulary and merge rules also matter.

4. Additional fourth corpus item

I added a fourth parallel text: a short question in English, Russian and Kazakh.

Tokenizer	English	Russian	Kazakh	RU/EN	KK/EN
o200k_base	12	17	20	$1.42\times$	$1.67\times$
cl100k_base	12	36	52	$3.00\times$	$4.33\times$

This additional example showed the same general pattern: the difference between languages depends strongly on the tokenizer.

5. Part 2 - Real API measurement

After the offline analysis, I moved to the real API experiment. I used the Google Gemini API with the model gemini-3.6-flash. The API key was kept in the .env file and was not included in the GitHub repository.

Language	Input	Output	Thinking	Total
English	101	72	382	555
Russian	131	102	566	799
Kazakh	237	180	632	1049

Compared with English, Russian used $1.30\times$ more input tokens and Kazakh used $2.35\times$ more input tokens. The Kazakh request also had more output and thinking tokens.

6. Part 3 - Cost calculation

For the calculation I used \$0.75 per 1 million input tokens and \$3.75 per 1 million output tokens. For billed output I included visible output and thinking tokens.

Language	Cost / request	Annual cost (5,000/day)
English	\$0.001778	\$3,245.31
Russian	\$0.002603	\$4,750.93
Kazakh	\$0.003223	\$5,881.52

At 5,000 requests per day there are 1,825,000 requests per year. The additional annual cost compared with English is about +\$1,505.63 for Russian and +\$2,636.21 for Kazakh. In this experiment, the total Kazakh cost was about $1.81\times$ the English cost.

The important point for me is that even when the meaning of the request is the same, the language can affect token usage and therefore the cost.

7. Task 2 - Two similar Kazakh sentences

Sentence	Chars	Bytes	Bytes/char	o200k	cl100k
Банк клиентке несие туралы хабар жасады.	40	74	1.85	11	21
Банк клиентке жаңа несие шартын түсіндірді.	43	80	1.86	14	30

The bytes per character were almost the same for both sentences, but the token counts were different. This showed me that tokenization is not determined only by UTF-8 encoding. The tokenizer also depends on which character combinations and words it can represent efficiently.

8. Production decision

For the API experiment, I tested gemini-3.6-flash. I would use it as a baseline model for a Kazakh support queue because it successfully handled the Kazakh requests in my experiment and I measured both token usage and cost. However, I would not make a final production decision from this experiment alone. I would first test several candidate models with the same Kazakh support questions and compare answer correctness, Kazakh-language understanding, response quality, token usage and cost.

9. Cost lever

One cost-control method that I did not use in this laboratory was prompt caching. It could be useful when the same system instructions or other prompt parts are repeated across many requests.

10. Conclusion

In this laboratory work, I learned that the price of an AI request depends not only on the meaning of the text, but also on how the tokenizer represents the language.

The actual results confirmed that Russian and Kazakh can require more tokens, but they also showed that bytes alone cannot predict token counts. The tokenizer itself is very important.

For my Gemini experiment, the Kazakh request had 237 input tokens compared with 101 for English. At 5,000 requests per day, the estimated annual cost was about \$5,881.52 for Kazakh versus \$3,245.31 for English. The main result for me was that language choice can directly affect token usage and API cost, even when the task and meaning are the same.